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ABSTRACT

The so-called D4 model is presented for the accurate computation of London dispersion interactions in density functional theory approximations (DFT-D4) and generally for atomistic modeling methods. In this successor to the DFT-D3 model, the atomic coordination-dependent dipole polarizabilities are scaled based on atomic partial charges which can be taken from various sources. For this purpose, a new charge-dependent parameter-economic scaling function is designed. Classical charges are obtained from an atomic electronegativity equilibration procedure for which efficient analytical derivatives with respect to nuclear positions are developed. A numerical Casimir-Polder integration of the atom-in-molecule dynamic polarizabilities then yields charge- and geometry-dependent dipole-dipole dispersion coefficients. Similar to the D3 model, the dynamic polarizabilities are precomputed by time-dependent DFT and all elements up to radon ($Z = 86$) are covered. The two-body dispersion energy expression has the usual sum-over-atom-pairs form and includes dipole-dipole as well as dipole-quadrupole interactions. For a benchmark set of 1225 molecular dipole-dipole dispersion coefficients, the D4 model achieves an unprecedented accuracy with a mean relative deviation of 3.8% compared to 4.7% for D3. In addition to the two-body part, three-body effects are described by an Axilrod-Teller-Muto term. A common many-body dispersion expansion was extensively tested, and an energy correction based on D4 polarizabilities is found to be advantageous for larger systems. Becke-Johnson-type damping parameters for DFT-D4 are determined for more than 60 common density functionals. For various standard energy benchmark sets, DFT-D4 slightly but consistently outperforms DFT-D3. Especially for metal containing systems, the introduced charge dependence of the dispersion coefficients improves thermochemical properties. We suggest (DFT-)D4 as a physically improved and more sophisticated dispersion model in place of DFT-D3 for DFT calculations as well as other low-cost approaches like semi-empirical models.

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I. INTRODUCTION

Many computational studies have shown that dispersion-corrected Kohn-Sham density functional theory (abbreviated as DFT in the following) is currently the method of choice for the routine computation of the electronic and geometric structure of large systems,^{1,2} e.g., in the fields of supramolecular chemistry,^{3–5} catalysis,^{6,7} or in materials science.^{8–10} In contrast to more elaborate wave function theory (WFT) based methods, most density functional approximations (DFAs) are not able to describe long-ranged

electron correlation effects.^{11–14} Their treatment is, however, important to compute energetic properties with high accuracy (approximately 1 kcal mol^{-1}), particularly for noncovalently bonded or condensed phase systems. Therefore, various correction schemes have been developed to describe so-called London dispersion interactions in a DFT framework.¹⁵

The most commonly used method for molecular applications is the so-called DFT-D3 scheme,¹⁶ which calculates the inter- and intramolecular dispersion interactions only by employing the given system coordinates (and atomic numbers). Similar atom pairwise

models, which additionally include information from the electron density, have been reviewed recently¹⁵ and are only mentioned briefly here. Among them are the exchange hole dipole moment^{14,17–19} (XDM) model and the Tkatchenko-Scheffler²⁰ (TS) model both employing Hirshfeld partitioning. It is also possible to directly develop nonlocal density functionals, which are inherently capable of describing London dispersion interactions. This way, atomic partitioning, which always involves some arbitrariness, is avoided. Here, the family of van der Waals density functionals is to be mentioned, which are based on the fundamental adiabatic connection theory and offer a rigorous basis for the design of dispersion-inclusive exchange-correlation functionals.^{21,22} A simplified construction scheme for the nonlocal correlation part has been introduced by Vydrov and Van Voorhis.^{23,24}

The computational costs of incorporating dispersion corrections into a standard DFT treatment are very method dependent but generally smaller compared to the actual DFT calculation. Consequently, it is reasonable to include them by default, and it has been demonstrated that the accuracy of dispersion-corrected DFAs for thermochemical properties on average follows Jacob's ladder²⁵ classification. Note that for very low-cost atomistic models such as semi-empirical molecular orbital or force-field methods, only non-density dependent schemes like DFT-D3 (and here proposed D4) are computationally feasible.

The current work describes the further development of the widely used semiclassical DFT-D3 approach. Recently, it was shown that this “geometry-only” model can be further improved by addition of atomic charge information.²⁶ Therein, we showed that computed atom-in-molecule dynamic polarizabilities can be scaled by means of an element specific function with Mulliken-type atomic charges as input.

Here, we want to report on the final version of the D4 model and provide it in a usable form for a large number of density functionals as DFT-D4. In general, we retained the general idea and strong points of the well established D3 scheme and introduced the charge dependence as well as some less important, mostly, technical improvements. Compared to the scaling scheme in Ref. 26, a less empirical function is used (see below) and the Mulliken partial charges are replaced by default with classical electronegativity equilibration (EEQ) partial charges as recently described by Goedecker *et al.*²⁷ The D4 scheme is, however, general in a sense that any type of atomic charges in addition to the geometric structure of the system can be used as input.

Previous studies have revealed that many-body dispersion (MBD) interactions beyond a pairwise picture are important, e.g., in supramolecular,^{28,29} in cluster,^{30,31} and in condensed-phase systems^{32–36} where so-called Dobson type B effects³⁷ play an important role. The use of a coupled-dipole based many-body dispersion (MBD) correction (cf. the MBD method in Refs. 38 and 39) in our implementation does not yield systematic improvements compared to the simple third-order Axilrod-Teller-Muto^{40,41} (ATM) term. However, the MBD correction proves to be useful for large systems (see noncovalent benchmarks in Sec. III), for which we recommend its application in terms of an energy correction.

The theory and technical details of the D4 method are described in Sec. II. Subsequently, in Sec. III, results for dispersion coefficients with D4 as well as for energies and structures obtained with DFT-D4 are compared directly to those of other established dispersion

correction schemes for the same underlying DFA. Finally, a summary and an outlook on possible future work will be given.

II. THEORY

Atomic units are used throughout in this work. The DFT-D4 workflow is simplified for an example molecule in Fig. 1. The pairwise dipole-dipole dispersion coefficients in D4 are calculated by numerical integration via the well-known Casimir-Polder relation

$$C_6^{AB} = \frac{3}{\pi} \int_0^\infty d\omega \alpha^A(i\omega) \alpha^B(i\omega). \quad (1)$$

At runtime, the isotropically averaged atomic dynamic polarizabilities at imaginary frequency $\alpha(i\omega)$ in Eq. (1) are obtained in two steps: The first step incorporates an atomic partial charge dependent scaling of atomic reference polarizabilities. For this purpose, a new charge-scaling function ζ has been designed in this work given by

$$\zeta(z^A, z^{A,ref}) = \exp \left[\beta_1 \left\{ 1 - \exp \left[\gamma^A \left(1 - \frac{z^{A,ref}}{z^A} \right) \right] \right\} \right], \quad (2)$$

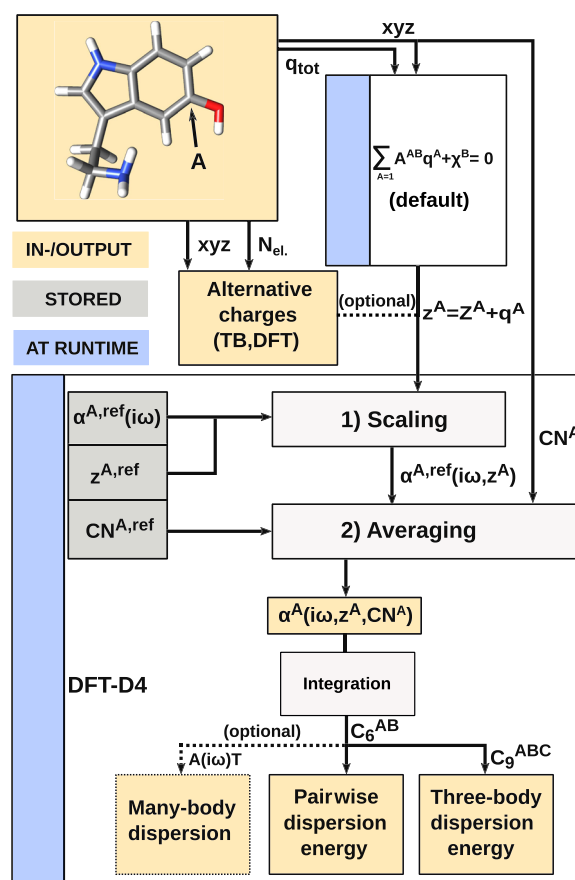


FIG. 1. Schematic workflow of the D4 program for an example molecule (all definitions and steps are explained in detail in the text).

which is sketched exemplarily in Fig. 2. In Eq. (2), the chemical hardness γ^A is taken from Ref. 42 and serves as a nonfitted element-specific parameter to control the steepness of the scaling function. The value of the global parameter $\beta_1 = 3$ was determined by inspection. The empirical charge-dependent scaling in D4 is intended to increase the magnitude of the atomic dynamic dipole polarizabilities $\alpha(i\omega)$ for larger number of electrons in proximity to the considered atom. Independent from the actual DFA which is to be corrected, environment dependent atomic partial charges are taken as input. The advantage is that such external partial charges can be (independently from the actual calculation) made accurate and robust and that the D4 dispersion coefficients remain functional independent. By default, classical EEQ type partial charges are used as descriptor for the electron density change (see Sec. II A). Other common options for the choice of the charges are discussed below.

In Eq. (2), we follow the definition of the effective nuclear charge z^A , used also in Ref. 26, as the sum of the nuclear charge of atom A and the atomic partial charge q^A

$$z^A = Z^A + q^A \quad (3)$$

and introduce effective nuclear charges for element specific reference systems $z^{A,ref}$. For all elements beyond krypton, we consistently employ modified nuclear charges Z' , as defined in Table I, due to the use of effective core potentials (ECPs) in the computation of the dynamic polarizability reference values.

In comparison to the initially proposed charge-scaling function presented in Ref. 26, the two element-specific parameters could be discarded resulting in a global scaling function with only one empirical parameter. The charge-dependent atomic dynamic polarizability for a single reference system of atom A is given by the product of $\alpha^{A,ref}(i\omega)$ and its scaling function as

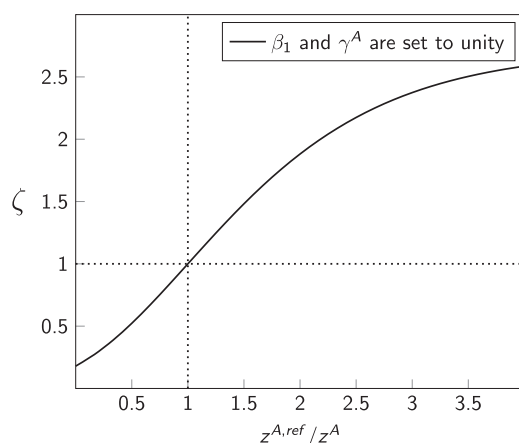


FIG. 2. The ζ -function is shown, which scales the time-dependent density functional theory (TD-DFT) computed dynamic polarizabilities $\alpha(i\omega)$ depending on the calculated effective nuclear charge z^A . The scaling depends on the quotient of the latter and the precalculated charge $z^{A,ref}$ of the atom in the reference system for which the dynamic polarizability has been computed. If both effective charges are equal, no scaling is performed (the crossing point of dotted lines) and the reference polarizability is taken.

TABLE I. Effective nuclear charges Z' used in the D4 charge scaling function for elements beyond krypton. The subtracted number corresponds to the number of core electrons absorbed in the ECP used (default def2-ECPs in TURBOMOLE^{43–45}).

Element	Z'
Rb–Xe	$Z-18$
Cs–La	$Z-46$
Ce–Lu	$Z-28$
Hf–Rn	$Z-60$

$$\alpha^{A,ref}(i\omega, z^A) = \alpha^{A,ref}(i\omega) \zeta(z^A, z^{A,ref}). \quad (4)$$

In Eq. (4), atom-in-molecule dynamic polarizabilities of element specific reference systems $\alpha^{A,ref}(i\omega)$ are utilized. Since such atom-in-molecule dynamic polarizabilities cannot be calculated directly, molecular dynamic polarizabilities of the reference systems $A_m X_n$ (having m chemically equivalent atoms A and n chemically equivalent X atoms) are used in the present model. From these molecular dynamic polarizabilities $\alpha^{A_m X_n}(i\omega)$, the contribution of the nX atoms is subtracted to obtain atom-in-molecule dynamic polarizabilities $\alpha^{A,ref}(i\omega)$ for atom A in different chemical environments. Here, the approximate additivity of polarizabilities⁴⁶ is exploited to generate atom-in-molecule dynamic polarizabilities according to the following partitioning scheme:

$$\alpha^{A,ref}(i\omega) = \frac{1}{m} \left[\alpha^{A_m X_n}(i\omega) - \frac{n}{l} \alpha^{X_l}(i\omega) \zeta(z^X, z^{X,ref}) \right]. \quad (5)$$

Equation (5) directly considers the charge scaling of all X atoms in the respective reference system. l is a stoichiometric factor specific to the reference molecule of element X . The effective nuclear charges $z^{X,ref}$ entering Eq. (5) are constant values determined once for the respective reference system. By subtracting charge-scaled polarizabilities of the atoms X , the partitioning of molecular polarizabilities into atomic contributions changes—with respect to D3—depending on the charge of the atoms X within each reference system. In D4, the polarizability of atom A is either increased (positive partial charge at X , subtract lower amount from A) or decreased (negative partial charge at X , subtract higher amount from A) when comparing with polarizabilities obtained by subtracting neutral X atoms.

The second step in the D4 procedure is the geometry-based interpolation over the charge-scaled element specific reference systems. In order to enable a geometric interpolation of all $\alpha^{A,ref}(i\omega, z^A)$ of atom A , a weighting procedure similar to the fractional coordination number (CN) based scheme in D3 is used. As already described in Ref. 26, the CN used within D4 is, however, slightly modified and includes an electronegativity difference dependence for the respective element-pair, i.e.,

$$\text{CN}^A = \sum_A \sum_{B \neq A} \frac{\delta_{AB}^{\text{EN}}}{2} \left(1 + \text{erf} \left(-k_0 \left(\frac{R_{AB} - R_{AB}^{\text{cov}}}{R_{AB}^{\text{cov}}} \right) \right) \right), \quad (6)$$

$$\delta_{AB}^{\text{EN}} = (k_1 \exp(|\text{EN}_A - \text{EN}_B| + k_2)^2) / k_3.$$

In Eq. (6), Pauling⁴⁷ electronegativities (EN), the internuclear distance R_{AB} of pair AB , and the covalent atomic radii⁴⁸ ($R_{AB}^{cov} = R_A^{cov} + R_B^{cov}$) are used. The parameters in Eq. (6) ($k_0 = 7.5$, $k_1 = 4.1$, $k_2 = 19.09$, and $k_3 = 254.56$) were obtained by fitting CN values to GFN2-xTB⁴⁹ derived Wiberg bond orders⁵⁰ of singly bonded diatomic molecules. The exponential used in D3 is replaced here by an error function to avoid a divergence behavior of the CN in applications for dense systems under periodic boundary conditions.⁵¹ A graphical comparison of CN values is shown for D4 and D3 in Fig. 3(a) for a catalyst frequently used in organometallic synthesis as an example.

All reference coordination numbers, termed $CN^{A,ref}$, are pre-calculated and stored in the code such that a Gaussian weighting function $W_A^{A,ref}$ can efficiently be used to generate system-specific charge- and geometry-dependent atom-in-molecule polarizabilities for atom A abbreviated as $\alpha^A(i\omega)$ for clarity

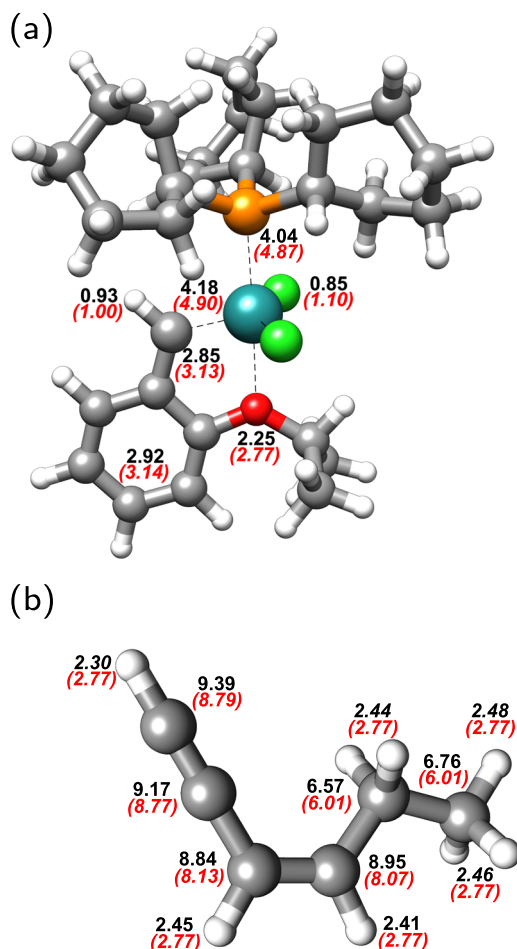


FIG. 3. (a) Coordination numbers of selected atoms for the first generation Hoveyda-Grubbs catalyst^{52,53} given as an example. Black/red (brackets) values show CN values for D4/D3. (b) Example molecule with depicted static atom-in-molecule polarizabilities $\alpha(0)$ (in bohr³) from D4 (black) according to Eq. (5) and for D3 (red in brackets). Values for carbon/hydrogen are given in bold/italic font.

$$\alpha^A(i\omega) \equiv \alpha^{A,ref}(i\omega, z^A, CN^A) = \sum_{A,ref=1}^{N^{A,ref}} \alpha^{A,ref}(i\omega, z^A) W_A^{A,ref}. \quad (7)$$

The contribution of each reference value $\alpha^{A,ref}(i\omega, z^A)$ to the final atom-in-molecule polarizability of atom A is given by

$$W_A^{A,ref}(CN^A, CN^{A,ref}) = \frac{\sum_{j=1}^{N^s} \exp(-\beta_2 \times j(CN^A - CN^{A,ref})^2)}{\sum_{A,ref=1}^{N^{A,ref}} \sum_{j=1}^{N^s} \exp(-\beta_2 \times j(CN^A - CN^{A,ref})^2)} \quad (8)$$

for $N^{A,ref}$ reference systems per element A (note that $\sum_{A,ref=1}^{N^{A,ref}} W_A^{A,ref} = 1$).

The parameter $\beta_2 = 6$ is adjusted manually to guarantee a smooth weighting function. In contrast to D3, the Gaussian weighting is changed such that several Gaussian functions can be used for single reference systems as shown in Eq. (8). Generally, the number of Gaussian functions N^s is obtained once for every reference system and used at runtime as described in Eq. (8). The procedure of setting N^s —for different reference systems—is exemplified in Fig. 4 to explain the principle of the weighting scheme. Differences between CN values of reference systems are used to indicate when another more compact Gaussian is necessary. As indicated in Fig. 4(a), reference systems with a sufficiently large CN difference can easily be distinguished from each other within the Gaussian weighting procedure in Eq. (8) for $N^s = 1$. This is no longer the case if the reference CN difference between two systems approaches smaller values as present between reference systems A, ref2 and A, ref3 (denoted as ΔCN_{23} in Fig. 4). The overlap between the Gaussian functions placed on A, ref2 and A, ref3—shown in white—makes it considerably more difficult to differentiate between both reference systems. For this reason, the set of Gaussian functions is enlarged—by increasing N^s —for these near lying reference systems, as shown in Fig. 4(b). This way, less overlapping functions are added, which makes both reference systems distinguishable within the Gaussian weighting procedure. By following this strategy, it is possible to generate a smooth weighting scheme without any discontinuities.

The final charge- and coordination-dependent atom-in-molecule polarizabilities, as obtained from Eq. (7), hence include the dependence of $\alpha(i\omega)$ on the spatially closest binding partners (i.e., so-called Dobson type A effects³⁷). They are used to calculate pairwise dispersion coefficients via a numerical Casimir-Polder integration over a fixed number of 23 points (between $i\omega_{min} = 10^{-6}i$ and $i\omega_{max} = 10.0i$ given in hartree)

$$\begin{aligned} C_6^{AB} &\equiv C_6^{AB}(CN^A, z^A, CN^B, z^B) \\ &= \frac{3}{2\pi} \sum_{j=1}^{22} (\omega_{j+1} - \omega_j) (\alpha^A(i\omega_{j+1}) \alpha^B(i\omega_{j+1}) + \alpha^A(i\omega_j) \alpha^B(i\omega_j)). \end{aligned} \quad (9)$$

The effect of charge scaling and Gaussian weighting on the $\alpha^A(0)$ values with variations in the partial charge (leading to different effective nuclear charges z) and in the coordination number is visualized for carbon and hydrogen in Fig. 5.

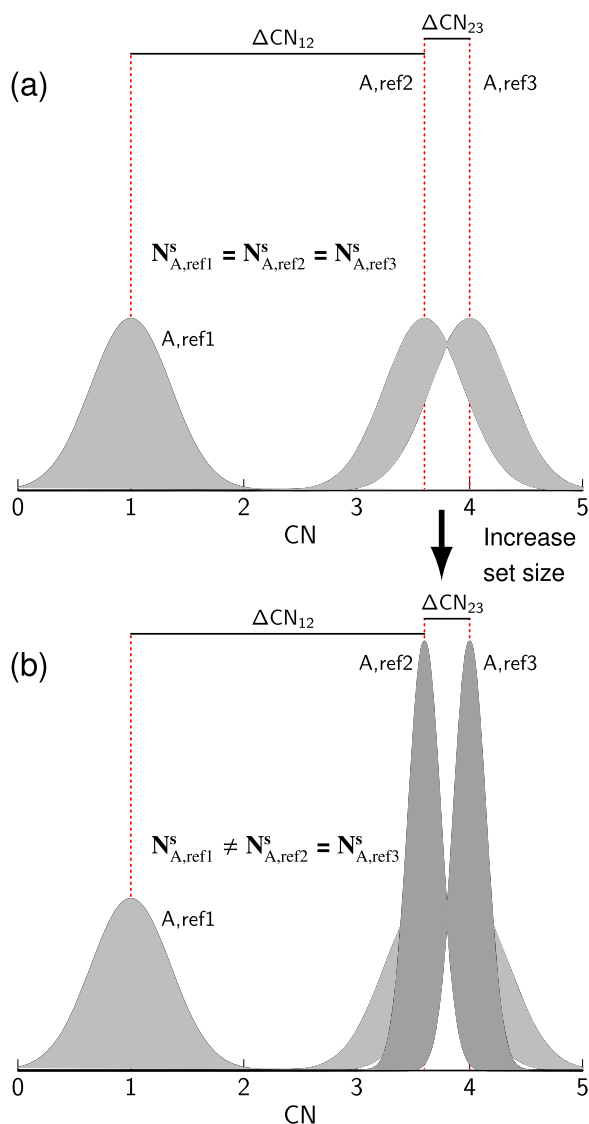
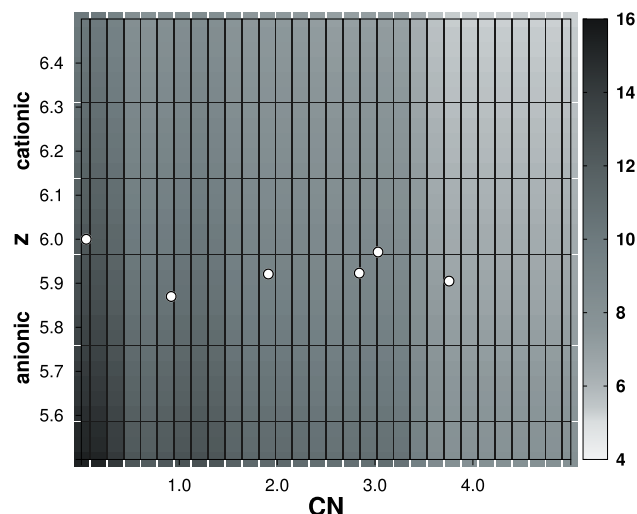


FIG. 4. Example for setting N^s for different reference systems. The CN difference between systems A, ref1 and A, ref2 is denoted as ΔCN_{12} . A, ref1 and A, ref2 are easily distinguishable within the Gaussian weighting procedure of Eq. (8) with $N^s = 1$, while A, ref2 and A, ref3 are not as could be seen in part (a) of the figure. To circumvent this behavior, the set of Gaussian functions is enlarged for A, ref2 and A, ref3 as it is shown in part (b) by varying the N^s value in Eq. (8) dynamically.

Furthermore, the lower part of Fig. 3 shows static atom-in-molecule polarizabilities for carbon atoms in different hybridization states for the (3Z)-hexen-1-yne molecule. Here, we follow the definition in Eq. (5) for D4 and the definition given in Ref. 16 for D3 to obtain those polarizabilities. As atomic partitioning schemes generally introduce some arbitrariness, the individual atom-in-molecule polarizabilities from D3 and D4 also differ (by about 10%). However, the physical observable, i.e., the total molecular dispersion coefficient, is similar with the two methods for this

(a) $\alpha^C(0)$ for carbon



(b) $\alpha^H(0)$ for hydrogen

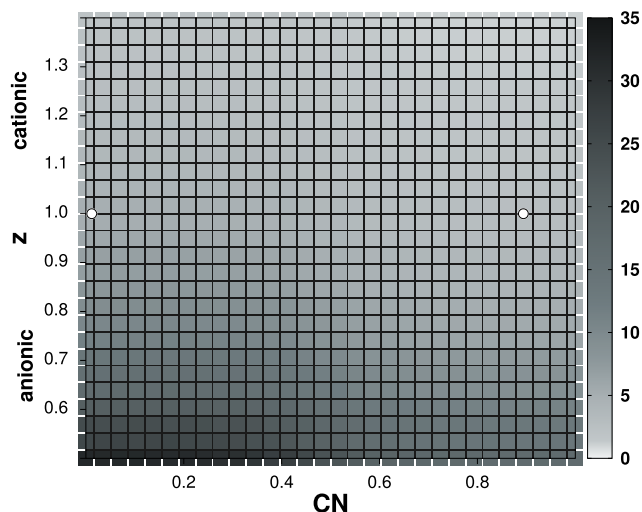


FIG. 5. Visualization of the two-dimensional dependence of static polarizabilities $\alpha(0)$ (in bohr³) on the charge and coordination state in the D4 model for (a) carbon and for (b) hydrogen. White circles represent $\alpha^A(0)$ values for the reference systems.

rather nonpolar compound [$C_{6,\text{mol}}^{AA}(\text{TD-PBE38/daug-def2-QZVP}) = 2103.1$, $C_{6,\text{mol}}^{AA}(\text{D4}) = 1949.5$, and $C_{6,\text{mol}}^{AA}(\text{D3}) = 1893.0$ —all given in Hartree bohr⁶]. Here, the additivity of pairwise dispersion coefficients in molecules P (atoms $p \in P$) and Q (atoms $q \in Q$) has been used

$$C_{6,\text{mol}}^{PQ} = \sum_p \sum_q C_6^{pq}. \quad (10)$$

A. Classical environment dependent partial charges

For the scaling of atom-in-molecule polarizabilities, an established classical charge model based on electronegativity equilibration of Gaussian type charge densities is used.²⁷ It allows the electronic charge to distribute itself in an optimal way over the whole system, includes penetration effects, and thus can describe both neutral and charged systems. Unlike environmental dependent partial charges that are determined by neural networks,⁵⁴ the adapted method determines the total charge (as sum of all atomic charges) of the system exactly. For this purpose, atomic charge densities are used within an isotropic electrostatics (IES) energy expression

$$E_{\text{IES}} = \mathbf{q}^T \left(\frac{1}{2} \mathbb{A} \mathbf{q} - \mathbf{X} \right), \quad (11)$$

where elements of the $\mathbf{X}$ vector and elements of the $\mathbb{A}$ matrix are given by

$$X^A = -\chi^A \quad \text{and} \quad A^{AB} = \begin{cases} J^{AA} + \frac{2\gamma^{AA}}{\sqrt{\pi}}, & A = B \\ \frac{\text{erf}(\gamma^{AB} R_{AB})}{R_{AB}}, & \text{otherwise.} \end{cases} \quad (12)$$

The EEQ model presented here contains as atomic radii a^A , which are used in the expression $\gamma^{AB} = ((a^A)^2 + (a^B)^2)^{-1/2}$, as well as element dependent atomic hardness terms J^{AA} , and the right-hand side (RHS) χ^A . The RHS consists of the atomic electronegativity EN^A which is scaled by the square-root of the error function modified D3 coordination number (termed mCN) that incorporates the environment dependency into the model including an element specific scaling parameter κ^A

$$\chi^A = \text{EN}^A - \kappa^A \sqrt{\text{mCN}^A}. \quad (13)$$

The mCN is given for atom A as

$$\text{mCN}^A = \frac{1}{2} \sum_{\substack{B=1 \\ B \neq A}} \left(1 + \text{erf} \left(-k \left(\frac{R_{AB}}{R_{AB}^{\text{cov}}} - 1 \right) \right) \right) \quad (14)$$

similar to Eq. (6) with $R_{AB}^{\text{cov}} = R_A^{\text{cov}} + R_B^{\text{cov}}$. Geometry-only dependent partial charges are obtained by solving a set of linear equations under the constraint that the atomic charges sum up to the correct overall charge

$$L = E_{\text{IES}} + \lambda \left(\sum_{A=1} q^A - q_{\text{tot}} \right), \quad (15)$$

with $\partial L / \partial \mathbf{q} = 0$ and $\partial L / \partial \lambda = \sum_{i=1}^N q_i - q_{\text{tot}}$. Adding this constraint in terms of a Lagrange multiplier leads to the modified linear system of equations

$$\begin{pmatrix} \mathbb{A} & \mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix} \begin{pmatrix} \mathbf{q} \\ \lambda \end{pmatrix} = \begin{pmatrix} \mathbf{X} \\ q_{\text{tot}} \end{pmatrix}. \quad (16)$$

This classical charge model requires five empirical parameters (J^{AA} , a^A , EN^A , κ^A , and R_{cov}^A) per element and achieves for molecules across

the entire periodic table of elements an average deviation of about $0.04e^-$ ($0.03e^-$ for organic molecules) from PBE0 based Hirshfeld charges (see the [supplementary material](#)). The main motivation to propose a classical charge model as default instead of a quantum chemistry based one is the higher robustness in electronically complicated cases for which simple tight-binding or DFT methods may fail to converge properly (see the [supplementary material](#) for discussion). Furthermore, it enables the use of D4 in combination with fast approaches.

For the construction of the analytical gradient of the D4 dispersion energy, the derivatives of the charges with respect to nuclear displacements are required. The partial derivative of the Lagrangian is derived with respect to internuclear distances in complete analogy to, e.g., coupled-perturbed SCF equations, and the analytical partial charge derivatives were developed in this work for the first time as given in the [supplementary material](#). Computer timings for the calculation of the charges and their derivatives are given in [Table II](#) for different protein structures. The formal scaling of the procedure is $\mathcal{O}(N^3)$ with number of atoms N and hence the same as for the dispersion energy in D3-ATM and D4 [for a two-body only approach—abbreviated by 2B—the formal scaling reduces to $\mathcal{O}(N^2)$].

By using the definition of the Lagrangian given in Eq. (15), the analytical charge gradients are derived as

$$\begin{pmatrix} \frac{\partial \mathbf{q}}{\partial \mathbf{R}_j} \\ \frac{\partial \lambda}{\partial \mathbf{R}_j} \end{pmatrix} = \begin{pmatrix} \mathbb{A} & \mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix}^{-1} \begin{bmatrix} -\left(\frac{\partial \mathbb{A}}{\partial \mathbf{R}_j} \right) \mathbf{q} + \left(\frac{\partial \mathbf{X}}{\partial \mathbf{R}_j} \right) \\ 0 \end{bmatrix}, \quad (17)$$

where the inverse of the indefinite $(N+1)$ matrix has been obtained by a Bunch-Kaufman factorization⁵⁵ and inversion.

B. Two-body dispersion energy

The pairwise dispersion coefficients are then used to compute the corresponding dispersion energy in complete analogy to DFT-D3 by multiplying with a short-range damping function in order to apply the model in combination with standard DFAs. The DFT-D4 pairwise dispersion energy is given by

TABLE II. Computer timings in seconds for the calculation of energies (E), energies and analytical gradients (E + g), and analytical charge derivatives ($\partial \mathbf{q} / \partial \mathbf{R}_j$) for differently sized protein structures with their protein database (PDB) entry. All calculations have been conducted at four cores [each CPU: Intel(R) Core(TM) i7-7700K CPU@4.20GHz] with DFT-D4-2B, DFT-D4-ATM, or DFT-D4-MBD. Timings for D4-MBD are excluded for proteins with more than 1500 atoms.

PDB (atoms)	E (D4)			E + g (D4)	
	2B	ATM	MBD	ATM	$\partial \mathbf{q} / \partial \mathbf{R}_j$
2BEG (373)	0.07	0.20	2.11	0.25	0.01
1R0I (782)	0.22	1.32	37.08	2.18	0.04
1MOL (1562)	1.13	9.75	401.00	13.28	0.40
2ZOH (1929)	1.80	18.21	...	24.85	1.29
1YMB (2489)	3.45	38.70	...	48.26	2.24
1JS8 (5988)	47.07	528.45	...	570.45	24.47

$$E_{\text{disp}}^{(6,8)} = - \sum_{AB} \sum_{n=6,8} s_n \frac{C_{AB}^{(n)}}{R_{AB}^{(n)}} f_{\text{damp}}^{(n)}(R_{AB}), \quad (18)$$

where s_n scales the individual multipolar contributions (s_6 and s_8 for the dipole-dipole and dipole-quadrupole term, respectively) and $f_{\text{damp}}^{(n)}$ denotes the rational Becke-Johnson (BJ) damping function [denoted as BJ-damping (BJD) in the following]. The damping function

$$f_{\text{BJD}}^{(n)}(R_{AB}) = \frac{R_{AB}^{(n)}}{R_{AB}^{(n)} + (a_1 R_0^{AB} + a_2)^{(n)}} \quad (19)$$

has become already the default in DFT-D3. For alternatives, see Ref. 56, and for a general discussion of damping functions in dispersion corrected DFT models, see Ref. 57. Equation (19) incorporates the functional-specific parameters a_1 and a_2 and the cutoff-radii defined as

$$R_0^{AB} = \sqrt{\frac{C_8^{AB}}{C_6^{AB}}}, \quad (20)$$

where the recursive relation between dipole-dipole and dipole-quadrupole dispersion coefficients is used as in DFT-D3(BJ).¹⁶ Furthermore, we define the following expression for the rational damping term:

$$R_{0,\text{BJ}}^{AB} = (a_1 R_0^{AB} + a_2). \quad (21)$$

C. Three-body dispersion and efficient geometry optimizations

The simplest way to include three-body effects is to use the well-known ATM term

$$E_{\text{ATM}}^{ABC} = \frac{C_9^{ABC} (3 \cos \theta_a \cos \theta_b \cos \theta_c + 1)}{(R_{AB} R_{BC} R_{CA})^3}. \quad (22)$$

Here, θ_a , θ_b , and θ_c are the internal angles of the triangle formed by R_{AB} , R_{BC} , and R_{CA} , respectively, while C_9^{ABC} is the triple-dipole constant given by

$$C_9^{ABC} = \frac{3}{\pi} \int_0^\infty d\omega \alpha^A(i\omega) \alpha^B(i\omega) \alpha^C(i\omega). \quad (23)$$

The numerical integration of the triple-dipole dispersion coefficient is possible using D4 polarizabilities, but due to the fact that the three-body energy contribution is rather small (at most 5%–10% of E_{disp}), the coefficients can be reasonably approximated as in the D3-ATM model by a geometric mean of dipole-dipole dispersion coefficients, i.e.,

$$C_9^{ABC} \approx \sqrt{C_6^{AB} C_6^{BC} C_6^{CA}}. \quad (24)$$

This approximation was already tested within the DFT-D3-ATM scheme for different element combinations, and a typically small deviation of about 10%–20% to the exact expression has been found.¹⁶ In the D4 model, the C_6 coefficients used to obtain C_9^{ABC} are obtained from charge-neutral atomic polarizabilities (i.e., neutral atoms with $z^A = z^B = z^C$). The finally used three-body dispersion energy expression is then given as

$$E_{\text{disp}}^{(9),\text{ATM}} = \sum_{ABC} E_{\text{ATM}}^{ABC} f_{\text{damp}}^{(9)}(\bar{R}_{ABC}), \quad (25)$$

where the sum is over all atom triples ABC applied with a zero-damping scheme proposed by Chai and Head-Gordon⁵⁶

$$f_{\text{damp}}^{(9)}(\bar{R}_{ABC}) = \frac{1}{1 + 6(\bar{R}_{ABC})^{-16}}. \quad (26)$$

Equation (26) includes the averaged interatomic distance

$$\bar{R}_{ABC} = \left(R_{AB} R_{BC} R_{CA} / R_{0,\text{BJ}}^{AB} R_{0,\text{BJ}}^{BC} R_{0,\text{BJ}}^{CA} \right)^{1/3}, \quad (27)$$

which incorporates $R_{0,\text{BJ}}^{AB/BC/CA}$ [cf. Eq. (21)]. The final energy expression used is therefore given as

$$E_{\text{disp}}^{\text{D4}} = E_{\text{disp}}^{(6,8)} + E_{\text{disp}}^{(9),\text{ATM}}. \quad (28)$$

Analytical gradients are available for this energy expression within the D4 implementation as tested against numerical derivatives.

D. Many-body dispersion energy

Depending on the size and the geometrical arrangement of the atoms, higher-order dispersion contributions (larger than three-body) can be of similar magnitude as three-body contributions and hence, for consistency, terms up to infinite order should be included to achieve a consistent description of all dipole-dipole interaction orders.³⁷ Here, a conceptually simple but robust approach introduced originally by Cao and Berne⁵⁸ is adapted, which has been made popular by Tkatchenko *et al.*³⁶ Physically, it is based on a coupling of atomic dipole polarizabilities in terms of quantum harmonic oscillators (QHOs). The coupled dipole model of QHOs serves as an approximation to describe the density-density response functions, which would otherwise be calculated, e.g., via the random phase approximation (RPA). The atomic response functions in a coupled dipole model allow a considerable reduction in the degrees of freedom (i.e., three QHOs per atom) and the computational costs. With this in mind, an alternative energy expression is proposed which consists of two parts. The first is to compose the two-body dipole-dipole and dipole-quadrupole interaction. The second part includes all dipole-dipole interactions up to infinite order $E_{\text{disp}}^{(n),\text{MBD}}$ ($n = 6, 9, 12, 15, \dots, \infty$). To avoid double counting of the two-body dipole-dipole energy, it is removed explicitly from the MBD energy according to

$$E_{\text{disp}}^{\text{D4-MBD}} = E_{\text{disp}}^{(6,8)} + (E_{\text{disp}}^{(n),\text{MBD}} - E_{\text{disp}}^{(6),\text{MBD}}). \quad (29)$$

Here, we have exploited that $E_{\text{disp}}^{(6)} = E_{\text{disp}}^{(6),\text{MBD}}$ (see the [supplementary material](#) for the complete derivation), which is the case in our consistent damping formulation for the two-body dispersion energy. It should be noted that this mutual consistency does not exist in the TS and MBD schemes of Tkatchenko and co-workers, which unfortunately has obfuscated the scientific discussion about the relevance of many-body dispersion beyond two- and three-body interactions. Furthermore, rearranging to $E_{\text{disp}}^{\text{D4-MBD}} = E_{\text{disp}}^{(n),\text{MBD}} + E_{\text{disp}}^{(8)}$ is not possible in the general case, as for double hybrid density functionals (abbreviated as DHDFs) $s_6 \neq 1$, whereas this scaling

cannot be applied to an individual term in the infinite-order MBD energy.

E. Definition of the D4 default model and use of alternative charges

At first, it should be emphasized that the D4 model only turns into a DFT-D4 method when used in combination with a specific density functional. If just polarizabilities or dispersion coefficients are calculated, the results are functional independent. The D4 default setting uses classical EEQ partial charges to scale atom-in-molecule polarizabilities due to their characteristics of being robust and efficient. The quality of the used charges is important but not essential for the finally obtained accuracy. The D4 model also works well with other charges, e.g., with partial charges obtained by the recently developed GFN2-xTB tight-binding method. This indicates its robustness and that under almost all circumstances D4 is similar or better than D3 but never worse (by construction). Generally, the use of three types of reference charges is currently implemented in the model: EEQ gas phase charges q (default), Mulliken-type partial charges q^{TB} from the GFN2-xTB tight-binding method,⁴⁹ and DFT Hirshfeld charges q^{DFT} (PBE0⁵⁹/def2-TZVP⁶⁰ level). Hirshfeld partial charges at the PBE0/def2-TZVP level of theory were also used to parameterize the EEQ model. If DFT charges should be used, they must be calculated separately such that they can be fed into the model. For a given type of charges (classical or TB or DFT), different reference effective nuclear charges $z^{A,ref}$, as introduced in Eq. (5), are stored and accordingly used in the D4 program.

The DFT-D4 default model always includes an ATM term as described in Sec. II C [see Eq. (28) for the total energy expression]. Nevertheless, we offer the possibility to use DFT-D4 including MBD effects (termed DFT-D4-MBD) in terms of an energy correction. The rational BJ-damping function is always applied. The use of other damping functions is not supported since we obtained in many test calculations the overall best results always with the BJ-damping for several DFA classes. When using this so-defined method, it should be abbreviated as “method-D4” (where “method” represents either a DFA or Hartree-Fock), which allows simple and clear referencing in future publications. If charges other than the default are used, this could be indicated by adding, e.g., “(TB)” or “(PBE0/def2-TZVP).” Unfortunately, the previous DFT-D3 method has partly lost a clear abbreviation over the years since additional parameterizations or extensions to the method have been assigned with various nomenclatures [see, e.g., DFT-D3(0),¹⁶ DFT-D3(BJ),⁵⁷ DFT-D3(CSO),^{61,62} DFT-D3(op),⁶³ and DFT-D3M⁶⁴].

In the next sections, the technical details of the calculations are given first (Sec. II F), followed by benchmarking of this finalized D4/DFT-D4 method for dispersion coefficients (Sec. III A), interaction energies (Sec. III B), conformational energies (Sec. III B), as well as in general thermochemical applications (Sec. III C). Finally, optimized covalent as well as noncovalent geometries are discussed (Sec. III D).

F. Technical details

1. Density functional theory calculations

All ground state DFT calculations were performed with either TURBOMOLE 7.0.2 and TURBOMOLE 7.2.1 (for all SCAN^{65,66}

calculations) or ORCA 4.0.1.^{67,68} Standard exchange-correlation energy integration grids (TURBOMOLE: *m4*, ORCA: *grid4*, *final-grid5*) and usual convergence criteria for the self-consistent field convergence (10^{-7} hartree) were used. The resolution of the identity (RI) approximation^{69–71} was applied in all calculations for the electronic Coulomb energy contribution. Ahlrichs’ type quadruple-zeta basis sets (def2-QZVP⁷²) were used throughout if not stated otherwise.

2. BJ-damping function parameterization

The functional specific parameters of the BJ-damping function have been determined using a Levenberg-Marquardt least-squares minimization to reference interaction energies of established non-covalent interaction (NCI) benchmark sets (S66x8,⁷³ S22x5,⁷⁴ and NCIBLIND10⁷⁵). In total, 98 dissociation curves with 718 reference data points of high accuracy were used for regression. The use of this new fitting set enabled parameterizations of such DFAs for which BJ-damping parameters could not be obtained successfully in earlier studies due to overbinding tendencies (e.g., Minnesota functionals; see Ref. 25).

3. Molecular dispersion coefficients

All molecular dynamic dipole polarizabilities $\alpha(i\omega)$ were calculated using time-dependent density functional theory (TD-DFT).^{76,77} As in D3, a variant of the PBE0 hybrid functional was used with a Fock-exchange admixture of 37.5% (dubbed PBE38). This method has already proved its accuracy and robustness in previous studies.^{16,26} The atomic orbital (AO) basis sets used in the TD-DFT calculations are of doubly augmented def2-QZVP quality very closely representing the complete basis set (CBS) limit for this property. Here, for the respective systems, each hydrogen has been augmented with additional (2s/2p), each main group element with additional (2s/2p/1d), and each transition metal with additional (2s/2p/1d/1f) Gaussian primitive functions. The exponent-extrapolation of the additional primitives was done with the subprogram DEFINE from TURBOMOLE 7.0.2. The following def2-ECPs are used: ECP-28 covering 28 core electrons (for Rb, Sr, Y–Cd, In–Sb, Te–Xe, and Ce–Lu), ECP-46 covering 46 core electrons (for Cs, Ba, and La), and ECP-60 covering 60 core electrons (for Hf–Hg, Tl–Bi, and Po–Rn), as defined in Ref. 60. Comparative calculations with the XDM model were conducted with the POSTG⁷⁸ program based on TURBOMOLE generated wave function *wfn*-files. Unfortunately, POSTG only supports a limited number of basis sets since each basis set has its own parameterization linked to the applied density functional. Therefore, all POSTG input files were calculated with the PBE⁷⁹/def2-TZVP setup. Furthermore, calculations were also carried out with the TS based MBD method of Tkatchenko *et al.* using the corresponding standalone code.⁸⁰ The required relative Hirshfeld volumes were taken from POSTG calculations performed at the PBE/def2-TZVP level.

All geometries of the TD-DFT derived organometallic molecular C_6 coefficient benchmark set (abbreviated as TOMC6 benchmark set) were obtained on the TPSS⁸¹-D3(BJ)-ATM/def2-TZVP level of theory, and molecular reference dispersion coefficients are shared in the [supplementary material](#).

4. Noncovalent reference interaction energies for L7 and S30L benchmark sets

We use recently published reference values (see the [supplementary material](#) of Ref. 82) for the L7⁸³ and the S30L⁸⁴ benchmark sets. They were obtained from local coupled-cluster single double perturbative triple excitation [CCSD(T)] calculations together with a special purpose CBS estimation scheme including the geometry deformation energy and Boys/Bernardi counterpoise (CP) correction⁸⁵ [DLPNO-CCSD(T)^{86,87} in its sparse matrix implementation⁸⁸ employing the CBS* protocol as described in Ref. 89].

5. Reference energies for thermochemical benchmarks

The reference conformational energies and structures of SCONF,⁹⁰ PCONF21,^{91,92} ICONF,²⁵ and UPU23⁸⁹ subsets were extracted from the GMTKN55 database (see Ref. 25 for further information and our homepage⁹³ for the entire database). For each benchmark set, the respective reference data with the accompanying level of theory are listed in the [supplementary material](#). The reference energies and structures for the MOR41 transition metal reaction benchmark set were taken from previous work (see Ref. 94 and our homepage⁹⁵ for the entire database).

6. Tetrakis(isonitrile)rhodium(I) dimer and monomer calculations

For the generation of reference association energies of the tetrakis(isonitrile)rhodium(I) complex, CP-corrected DLPNO-CCSD(T) calculations with tight thresholds and extended basis sets

[def2-TZVPP(VeryTightPNO⁹⁶)/def2-QZVPP(TightPNO⁹⁶)] were conducted using ORCA 4.0.1. A basis set extrapolation was performed using optimized exponents proposed by Neese and Valeev.⁹⁷ The deformation energy of the monomers (0.42 kcal mol⁻¹) was also taken into account. The error bar of the calculated interaction energy is estimated to about ± 0.5 kcal mol⁻¹. Geometries have been obtained at the PBEh-3c⁹⁸ level of theory and verified as minimum structures by frequency calculations.

III. RESULTS

A. Molecular dispersion coefficients

We have taken reference molecular dispersion coefficients which were determined experimentally from dipole oscillator strength distributions (DOSDs) as described in previous studies by Meath and co-workers.^{99,100} From this data, a benchmark set has been compiled in Ref. 20 consisting of 1225 molecular dispersion coefficients for systems ranging from dihydrogen to octane and other nonorganic molecules such as SF₆ or O₂. The respective statistical evaluation for D3,¹⁰¹ D4, D4(TB), the local response dispersion method (LRD),¹⁰² and TS²⁰ for all systems in the DOSD benchmark set is given in Table III. For medium-sized organic and inorganic molecules, D4/D4(TB) further improves upon the already accurate D3 model (mean absolute deviation, MAD, of 3.8% and 3.9% vs 4.7%). Other statistical measures for D4/D4(TB) are also lowered in comparison to values for the competitors which indicates a robust and consistent improvement in the description of pairwise dispersion coefficients for such systems. In the authors' opinion, the

TABLE III. Top (entries labeled for DOSD set): Statistical measures for the relative deviation (in percentage) of calculated molecular C₆ dispersion coefficients for different approaches with respect to experimental values for a molecular benchmark set consisting of 50 molecules. For D3, LRD, and TS, values are taken from Refs. 20, 101, and 102, respectively. Bottom (entries labeled for TOMC6 set): Statistical measures for the relative deviation (in percentage) of calculated homomolecular C₆ dispersion coefficients for different approaches with respect to theoretical TD-DFT values for a transition metal benchmark set of 25 complexes for the complete set and 15 complexes for the subset. The best values are highlighted in bold font.

		Measure	D4	D4(TB)	D3	LRD	TS
DOSD		MAD	3.8	3.9	4.7	6.1	5.3
		MD	−0.1	0.3	2.4	−2.5	−2.7
		SD	5.1	5.3	5.2	7.7	7.3
		AMAX	29.1	34.7	23.9	52.9	44.0
			D4	D4(TB)	D3	XDM	MBD
TOMC6	Subset	MAD	8.7	8.9	20.5	9.4	13.7
		MD	−1.7	−2.3	−9.6	1.4	13.7
		SD	11.2	11.4	28.1	13.3	6.1
		AMAX	29.7	25.1	80.8	36.6	26.2
		Measure	D4	D4(TB)	D3		
	Complete	MAD	7.0	7.9	12.0		
		MD	−2.2	−3.6	−5.5		
		SD	9.3	9.4	15.7		
		AMAX	29.7	25.1	56.1		

achieved MAD of 3.8% closely approaches the inherent accuracy of the underlying TD-DFT calculations and is thus difficult to improve further. For comparison, an assessment of the XDM method yielded an MAD of 10.0% for a similar small-molecule database,¹⁹ while a number of vdW density functionals yield even larger deviations for asymptotic molecular C_6 coefficients.¹⁰³ Note that the resulting DFT-D4 interaction energy errors in the asymptotic regime of about 4.0% are comparable or even smaller than, e.g., residual errors in WFT energy calculations employing large but finite triple- or quadruple-zeta basis sets.

Since the DOSD set excludes the important class of transition metal complexes, we have created a corresponding benchmark set consisting of computed molecular dipole-dipole dispersion coefficients derived from TD-DFT dynamic reference polarizabilities at the PBE38/daug-def2-QZVP level of theory. This benchmark set consists of 25 organometallic complexes (see the upper part of Fig. 6 for example structures) and is dubbed as TOMC6 which stands for the TD-DFT derived organometallic molecular C_6 coefficient benchmark set. Results for the D3, D4, D4(TB), XDM, and MBD schemes are given in the lower part of Fig. 6 together with their statistical evaluation listed in Table III. Here, the TOMC6 set was divided into two parts because several systems could not be treated with XDM and MBD for technical reasons. The resulting smaller set for which XDM and MBD data are also available is termed “subset” in the following. In accordance with the previously discussed DOSD benchmark set, the use of scaled polarizabilities in D4/D4(TB) improves upon the accuracy of the D3 method. For this set, only XDM is able to achieve a comparable accuracy, however, at a substantially larger computational effort because a properly converged DFT electron density is required in XDM. Furthermore, XDM energies need to be integrated numerically over a fine grid to avoid numerical noise. The MBD model achieves a result which is in between of D4/D4(TB) and D3 (here, the molecular dispersion coefficient changes for different range-separation parameters—we used $\beta^{PBE} = 0.83$ —and, therefore, for different DFAs within MBD). Notably, all methods yield larger errors for the electronically more complicated systems in the TOMC6 (sub)set than for the DOSD molecules. Nevertheless, the asymptotic D4 error of about 7.0% is still smaller than typical errors of DFAs for other correlation energy effects. This will be further discussed below for the thermochemistry of transition metal complexes. The application of Hirshfeld partial charges at the PBE0/def2-TZVP level of theory slightly worsens upon the D4(TB) result with an MAD of 9.5% for the complete set. The fact that calculated D4 molecular dispersion coefficients have the smallest errors with respect to reference values when applying the EEQ charges additionally motivates their use as default (as already defined in Sec. II E). An explicit discussion of the effect of using different charges is avoided, whereby DFT-D4 results with GFN2-xTB charges are shared in the [supplementary material](#) for interaction energies and structures.

B. Noncovalent interactions and conformational energies

As shown in Sec. III A, the pairwise dispersion coefficients for organic/inorganic molecules and transition metal complexes are improved with the D4 method. In general, it is assumed that improved dispersion coefficients in DFT-D type methods are

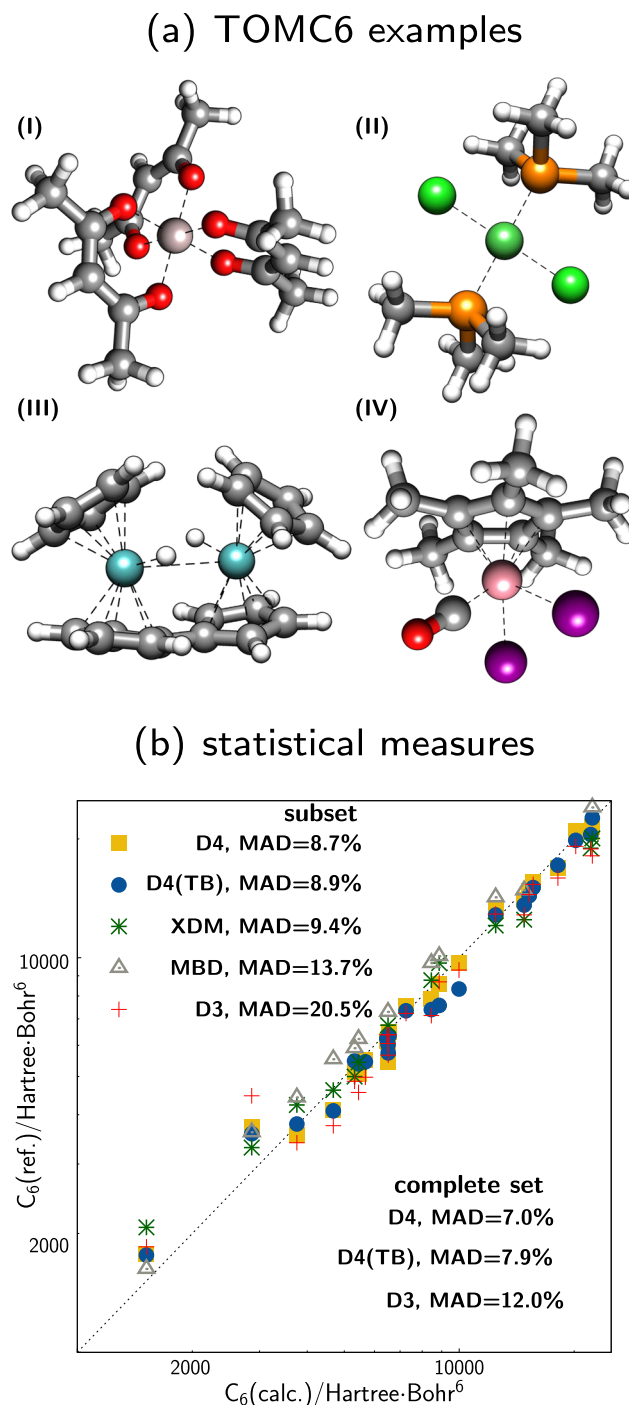


FIG. 6. (a) Example complexes from the TOMC6 set including (I) aluminum, (II) nickel, (III) molybdenum, and (IV) cobalt to demonstrate the diversity of the benchmark set. Structures and reference molecular dispersion coefficients are listed in the [supplementary material](#). (b) Correlation plot between reference molecular dispersion coefficients of transition metal complexes obtained by hybrid TD-DFT [numerically integrated from molecular $\alpha(i\omega)$ values according to Eq. (9)] and molecular ones derived from pairwise dispersion coefficients with D3, D4, D4(TB), XDM, and MBD models.

associated with improved noncovalent interaction energies. This assumption is to be verified in this section for various small to large molecule benchmark sets. Reference interaction energies refer to the CCSD(T) or DLPNO-CCSD(T) level of theory with tight threshold settings and CBS extrapolation mostly taken from Ref. 82. The upper part of Fig. 7 shows interaction energies in benchmark sets of increasing molecule size which are mainly stabilized by London dispersion: S22,¹⁰⁵ L7,⁸³ and S30L.⁸⁴ Various typical noncovalent interaction motifs such as hydrogen and halogen bonding, π - π stacking, nonpolar dispersion, CH- π , and cation-dipolar interactions are represented.

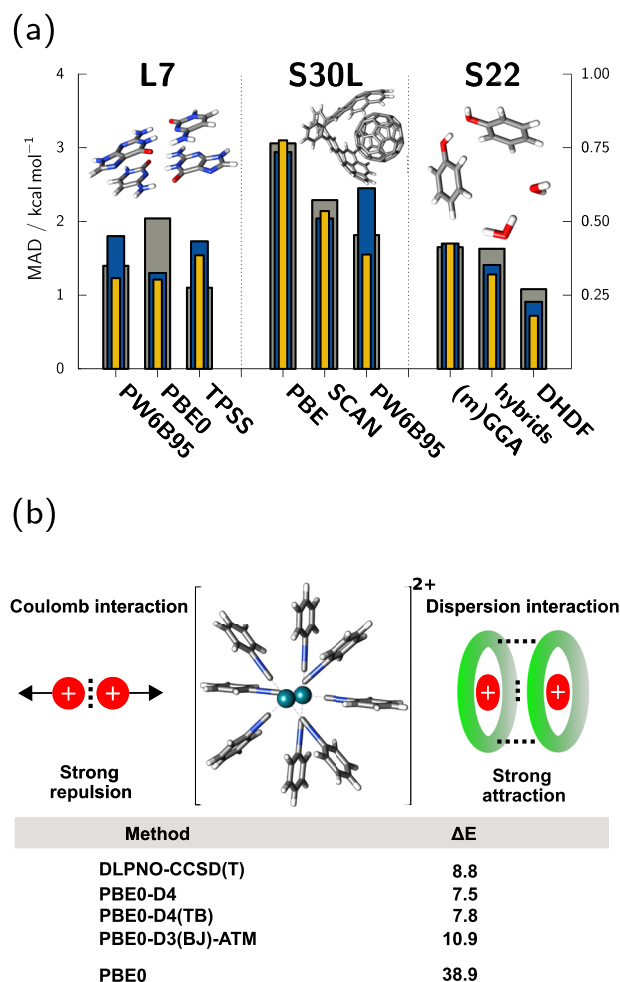


FIG. 7. (a) Comparison of MAD values for three typical noncovalent interaction benchmark sets (L7, S30L, and S22) and various standard DFAs. Density functionals corrected by D3(BJ)-ATM are shown in gray (bar width 1.0), DFT-D4 results are shown in blue (bar width 0.4), and DFT-D4-MBD results are shown in yellow (bar width 0.2). For the S22 benchmark set, the scaling of the MAD axis was adjusted accordingly and values were obtained by averaging over several typical DFAs (eight *meta*-GGAs, nine hybrids, and three DHDFs). (b) Shown is a cationic dirhodium complex with the respective association energy ΔE in the gas phase where the repulsive cation-cation interaction is almost compensated by attractive London dispersion interactions, schematic adapted from Ref. 104.

The description of noncovalent interactions in the larger systems is slightly (S30L) to largely (L7) improved at the DFT-D4-MBD level compared to the already well performing DFT-D3(BJ)-ATM method. The only exception is the TPSS-D4-MBD treatment for the L7 set which is worse than the D3(BJ)-ATM corrected values. Note that residual MAD values of about 1 kcal mol⁻¹ for L7 and 1-2 kcal mol⁻¹ for S30L are not far from the accuracy of the reference data. For the S30L benchmark, the MAD values increase when the ATM term is neglected, i.e., for DFT-D3 [DFA/MAD(DFA-D3(BJ))]/MAD(DFA-D3(BJ)-ATM), in kcal mol⁻¹: PW6B95/3.4/2.4, TPSS/4.1/3.6, B3LYP/5.9/4.1, PBE/2.3/2.8. For the S22 set, we use averaged MAD values for different DFA classes arranged according to Jacob's ladder. Data for eight (meta) generalized gradient approximation DFAs, abbreviated as (m)GGA (namely, BLYP,^{106,107} BP86,¹⁰⁸ M06L,^{109,110} O-LYP,^{111,112} PBE, revPBE,¹¹³ RPBE,¹¹⁴ and SCAN), nine hybrid DFAs (M06,¹¹⁵ B3LYP,^{116,117} B3LYP, M062X, O3LYP, PBE0, PW6B95,¹¹⁸ TPSS0, and TPSSH), and three DHDFs (DSD-BLYP,¹¹⁹ DSD-PBEB95,¹²⁰ and PWPB95¹²⁰) are used to obtain averaged MAD values for S22. For this important and prototypical noncovalent interaction (NCI) benchmark, D4 outperforms D3(BJ)-ATM especially for hybrids and DHDFs.

The lower part of Fig. 7 shows an interesting example for strong London dispersion interactions in a doubly positively charged organometallic dirhodium complex. Here, the dominant Coulomb repulsion can almost be compensated by dispersion interactions¹⁰⁴ in the gas phase [compare the association energy ΔE (PBE0) in Fig. 7(b) with its corresponding dispersion corrected values]. This and related complexes have been studied intensively, and the property of oligomer formation in a solution has been an "open case" for many years.¹²¹⁻¹²³ Quantum chemical calculations have been able to assign the main binding motif of this highly charged system to be the London dispersion thus making it an ideal test case for the DFT-D4 method. The theoretical reference association energy was calculated with an accurate local coupled-cluster protocol (see Sec II F) and is compared to the dispersion and Boys-Bernardi CP-corrected PBE0/def2-QZVPP values.

Counterintuitively, the D4 corrected gas phase interaction is more favorable (less positive association energy) than the corresponding D3 value. This is initially surprising since cationic compared to neutral complexes should feature smaller dispersion coefficients as accounted for in D4 but not in D3. According to a detailed analysis, the observed increased dispersion interaction in DFT-D4 is based on new reference systems that have been added to some elements within the D4 model (e.g., RhH₅, CN^{Rh} = 4.70 and C₆H₆, CN^C = 2.92), increasing the homo-atomic dispersion coefficient for those elements {e.g., for rhodium atoms within the complex, the difference between D3 [C₆^{AA}(D3) = 244.7 hartree bohr⁶] and D4 [(C₆^{AA}(D4) = 294.9 hartree bohr⁶] is significant}. Furthermore, the molecular charge is distributed over all atoms in the molecule and is not centered on the rhodium atoms. Thus, the partial charges of the rhodium atoms are only marginally changed when the neutral dimer complex is turned into the dication (the change in q^{Rh} is less than 0.1). In summary, the stronger binding in D4 which is in better agreement with the localized CCSD(T) reference value (for both types of partial charges—EEQ and TB) can be explained by larger atomic dispersion coefficients in combination with only a

small decrease in atomic polarizabilities due to small partial charges on the central rhodium atoms.

DFT-D4 also yields comparable or higher accuracies for (bio)chemically important conformational energies in various test systems. Figure 8 exemplifies this for sugar conformers (SCONF)

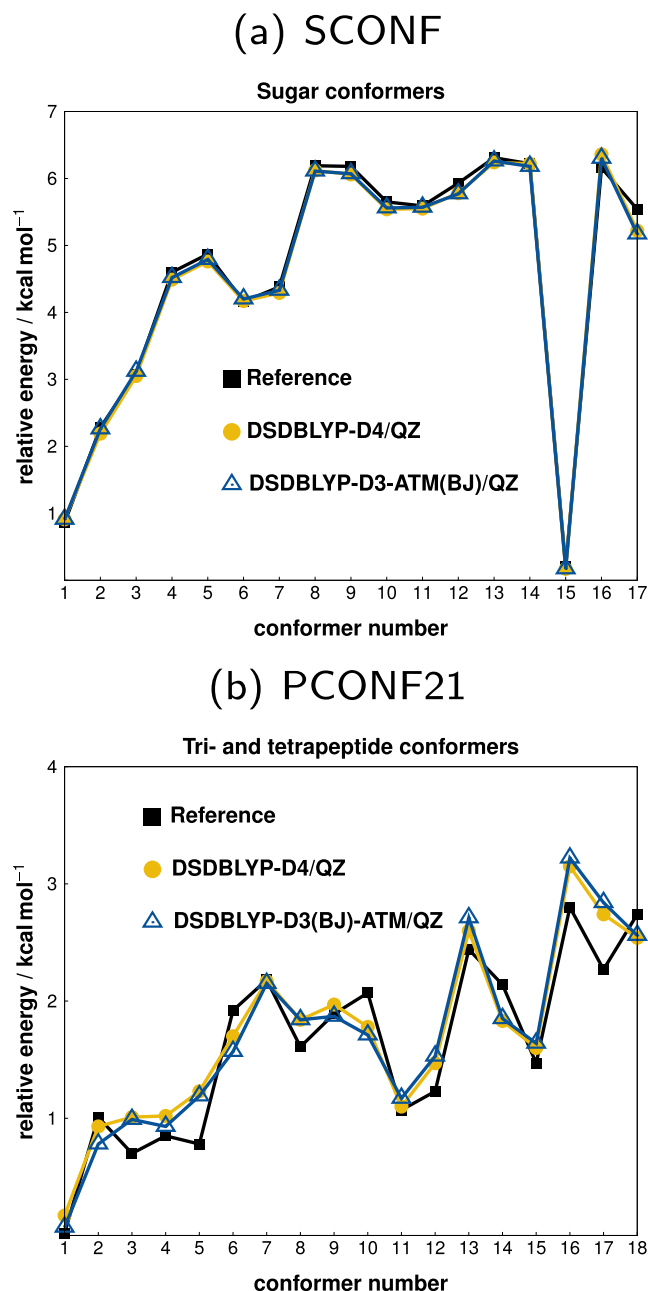


FIG. 8. Conformational energies for two benchmark sets (SCONF and PCONF21) in comparison to accurate reference values at the DLPNO-CCSD(T)/TightPNO/CBS level of theory.²⁵ For all calculations, a def2-QZVP basis set (abbreviated as QZ) has been applied.

as well as for tri- and tetrapeptide conformers (PCONF21) both taken from the GMTKN55 database. Additionally, results for conformational energies of inorganic molecules (ICONF) as well as for RNA backbone conformers (UPU23) are given in Table IV. Note that in contrast to DFT-D3, the DFT-D4 training set used for the determination of the BJ-damping parameters does not contain conformational energies. In Fig. 8, we show DSD-BLYP based results as an example and provide statistical evaluations for other selected DFAs (PBE, B3LYP, and PW6B95) in Table IV.

Overall, the conformational energies are generally improved for PBE, B3LYP, and PW6B95 by applying DFT-D4/DFT-D4(TB) compared to DFT-D3(BJ)-ATM. The reduction of the MAD is often noticeable (about 0.1 kcal mol⁻¹) but sometimes significant, e.g., 0.20–0.25 kcal mol⁻¹ for peptides and RNA structures with PBE or B3LYP. Especially, for the peptide set, the improvement is large when compared to the small averaged absolute relative energy of only 1.62 kcal mol⁻¹.

C. Thermochemistry

As discussed in Sec. III A, the D4 model is particularly good for systems in which atomic partial charges show significant deviations from the reference systems used in the Gaussian weighting procedure [cf. Eq. (8)]. Typical examples are transition metal complexes, which contain *d*-block elements in varying oxidation states. We have discussed the improved description of pairwise dispersion coefficients for transition metal complexes for the TOMC6 set (see Sec. III A). In the following, the thermochemistry of transition metal compounds is investigated employing the recently composed MOR41 set consisting of 41 closed-shell transition metal reactions of uncharged molecules. The estimated maximum error of the reference reaction energies is about 2 kcal mol⁻¹ (see Ref. 94 for further information). The reactions in this set typically occur in homogeneous catalysis, i.e., complexation reactions, oxidative additions, and ligand exchange reactions.

For the MOR41 set, the D4 model outperforms D3(BJ)-ATM for all tested DFAs (the only exception is PW6B95-D4). This is particularly noticeable for the well performing double hybrid DFAs, DOD-PBE¹²⁴ and DSD-PBE,¹²⁴ or hybrid DFAs (especially for PBE0) for which it is now possible to almost reach the estimated error level of the reference method. Hence, the combination of such DFAs with D4 represents an efficient route to obtain reaction energies in transition metal thermochemistry applications. During the D4 parameterization process, D3 damping parameters for some DFAs could also be “repaired.” The RPBE functional is given as an example, where the published parameters were apparently not optimal (see the supplementary material of Ref. 25 for original values). A recently published work shows furthermore that DFT-D4 is particularly suitable for the description of noncovalent interactions in organometallic complexes.¹²⁵

For main group elements, the general performance of DFT-D4 for thermochemistry and kinetics is assessed using the large GMTKN55 database. This database allows an evaluation for a wide variety of chemical problems and provides a large number of 2462 systems resulting in 1505 relative energies. The lower part of Fig. 9 shows results for several dispersion-corrected DFAs grouped according to Jacob’s ladder plotted with their weighted mean total absolute deviations (WTMAD-2; see Ref. 25 for the exact

TABLE IV. Conformational energies calculated with three widely used dispersion corrected DFAs (PBE, B3LYP, and PW6B95) with the def2-QZVP basis set. All statistical measures are given in kcal mol^{−1} relative to the reference energies. The best result for each measure is highlighted in bold, DFT-D3(BJ)-ATM is abbreviated by D3, DFT-D4 is abbreviated by D4, and DFT-D4(TB) by D4(TB). The average reference energy over all entries in the set $|\overline{\Delta E}|$ is taken from Ref. 25.

Sugar conformers (SCONF, $ \overline{\Delta E} = 4.60$)									
Measure	PBE			B3LYP			PW6B95		
	D3	D4	D4(TB)	D3	D4	D4(TB)	D3	D4	D4(TB)
MAD	0.78	0.88	0.77	0.29	0.33	0.23	0.24	0.28	0.20
MD	0.28	0.34	0.29	−0.14	− 0.07	−0.14	− 0.01	0.04	−0.02
SD	0.98	1.09	0.94	0.48	0.58	0.42	0.38	0.43	0.36
AMAX	2.73	2.96	2.67	1.73	2.07	1.64	1.21	1.33	1.12
Tri- and tetrapeptides (PCONF21, $ \overline{\Delta E} = 1.62$)									
MAD	1.20	0.99	1.04	0.53	0.35	0.30	0.48	0.49	0.49
MD	−0.54	− 0.51	−0.53	−0.04	0.03	0.01	0.34	0.36	0.38
SD	1.44	1.24	1.18	0.60	0.41	0.37	0.58	0.60	0.46
AMAX	2.51	2.31	2.36	1.11	0.90	0.88	1.05	0.98	0.98
Inorganic conformers (ICONF, $ \overline{\Delta E} = 3.27$)									
MAD	0.31	0.31	0.31	0.28	0.28	0.27	0.22	0.22	0.22
MD	0.09	0.09	0.10	− 0.06	−0.08	−0.07	0.04	0.02	0.02
SD	0.38	0.40	0.40	0.39	0.42	0.39	0.33	0.33	0.34
AMAX	0.84	1.12	1.11	0.99	1.15	0.98	0.90	0.83	0.85
RNA backbone conformers (UPU23, $ \overline{\Delta E} = 5.72$)									
MAD	0.57	0.51	0.48	0.68	0.60	0.54	0.66	0.63	0.59
MD	0.36	0.27	0.20	0.55	0.42	0.31	0.52	0.49	0.42
SD	0.71	0.63	0.58	0.80	0.70	0.57	0.80	0.76	0.58
AMAX	1.67	1.62	1.50	1.57	1.50	1.33	1.65	1.55	1.44

definition). The figure is divided into two parts describing the performance for the entire GMTKN55 database (termed GMTKN55 full) and for all its NCI subsets (shown inverted and termed GMTKN55 NCIs).

Overall, for this typical selection of DFAs, DFT-D4 represents a retention or small improvement over DFT-D3(BJ)-ATM for general thermochemistry. No single outlier in the huge number of systems in GMTKN55 was detected. The improvement obtained with DFT-D4 is more pronounced in the NCI subsets than for the entire database because in the former, the dispersion contributions are relatively larger than for most “normal” chemical reactions. A direct comparison between DFT-D4 (ATM, MBD either with EEQ or TB charges) and nonlocally corrected DFT (DFT-NL) for this NCI subsets shows a clear improvement when applying D4 (see Table V).

Importantly, the remaining MAD of 2.0 kcal mol^{−1} for the MOR41 set and the WTMAD-2 of 2.0–3.0 kcal mol^{−1} for GMTKN55 with the best functionals is not far from the accuracy of the underlying reference methods.

The s_6 damping parameters for the double hybrid DFAs (which is unity for all lower-rung DFAs) have been constrained in the fitting

according to the procedure of Ref. 16, i.e., we have not followed the construction scheme of Martin and co-workers¹²⁴ for its determination. This constraint, however, does not affect their accuracy which is actually significantly increased with DFT-D4 compared to the already very accurate DFT-D3 versions. Overall, the performance of D4 and D4-MBD is very similar for the whole GMTKN55 benchmark database which shows that an ATM treatment is sufficient if the damping function is chosen in a consistent manner (see Secs. II C and II D).

D. Covalent and noncovalent structures

In order to investigate the DFT-D4 accuracy more closely, various standard benchmark sets for bond distances in equilibrium geometries are discussed here. This is of particular importance because structure determination is a common application area of DFT methods. First, covalent bond lengths are to be investigated, whereby noncovalent distances and entire equilibrium structures are also examined in this section. All geometries discussed here are fully optimized employing an Ahlrichs' type quadruple-zeta AO basis set (def2-QZVP⁷⁴).

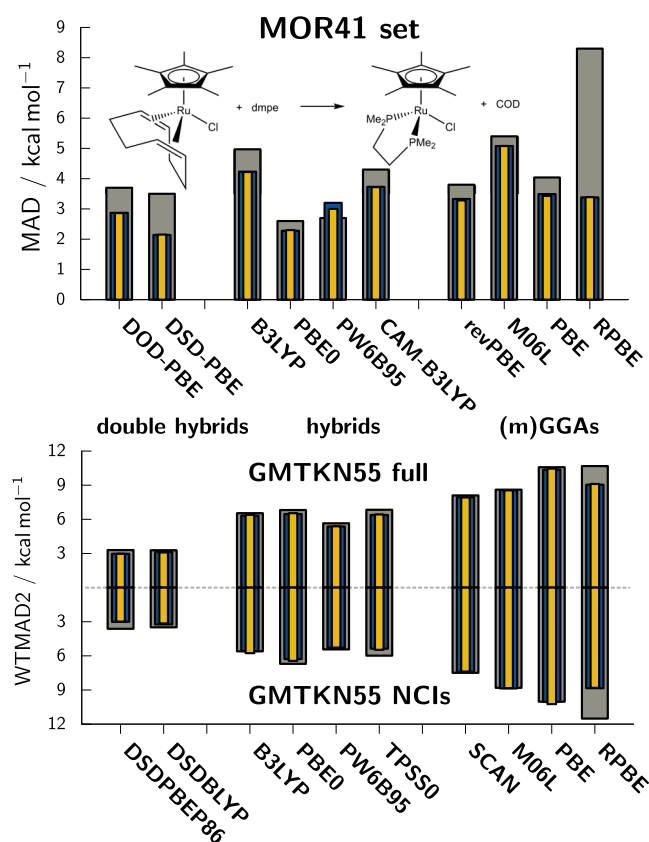


FIG. 9. (Upper) Comparison of MAD values for MOR41 reaction energies. The inset shows one example reaction (reaction 28 in the set). Various density functionals from different rungs of Jacob's ladder are investigated. Shown are deviations of D3(BJ)-ATM corrected (gray, bar width 1.0), D4 corrected (blue, bar width 0.4), and D4-MBD corrected (yellow, bar width 0.2) DFT computed reaction energies to reference values with an estimated uncertainty of ≈ 2 kcal mol $^{-1}$. (Lower) Comparison of weighted mean deviations (WTMAD-2 values) for the entire GMTKN55 set and its noncovalent interaction subsets. Shown are deviations of DFT-D3(BJ)-ATM (gray), DFT-D4 (blue), and DFT-D4-MBD (yellow) values to reference data (see Sec II F for further information).

The sets used for testing covalent bond lengths consist of molecules formed by the first or second row main group elements of the periodic table (LMGB35), molecules which are composed of main group elements of the third or higher rows (HMGB11),

TABLE V. WTMAD-2 values over all NCI subsets of the GMTKN55 benchmark database given in kcal mol $^{-1}$ for DFT-D4 [(EEQ/TB)-ATM, (EEQ/TB)-MBD] or DFT-NL which were extracted from Ref. 126.

DFA	EEQ		TB		DFT-NL
	ATM	MBD	ATM	MBD	
B3LYP	5.58	5.76	6.29	6.36	6.03
PW6B95	5.29	5.27	5.38	5.50	5.87
DSD-BLYP	3.23	3.15	3.09	3.12	3.18

and 3d-transition metal complexes with a total of 50 analyzed bond lengths as compiled by Bühl and Kabrede¹²⁷ (TMC32). For a detailed description of the former benchmark sets, see Ref. 98. These sets are ideally suited also for “cross-checking” since many of the here treated elements were not part of the D4 parameterization process (only the elements H, C, N, and O are present in the BJ-damping function fit), which was furthermore based solely on energies.

Figure 10(a) shows mean absolute deviations from reference values for three typical D3(BJ)-ATM and D4 dispersion corrected DFAs, and a statistical evaluation of all three benchmark sets is given in the supplementary material.

Inspection of Fig. 10(a) shows only marginal differences between DFT-D4 and DFT-D3(BJ)-ATM for bond lengths which is expected because dispersion corrections mostly affect the

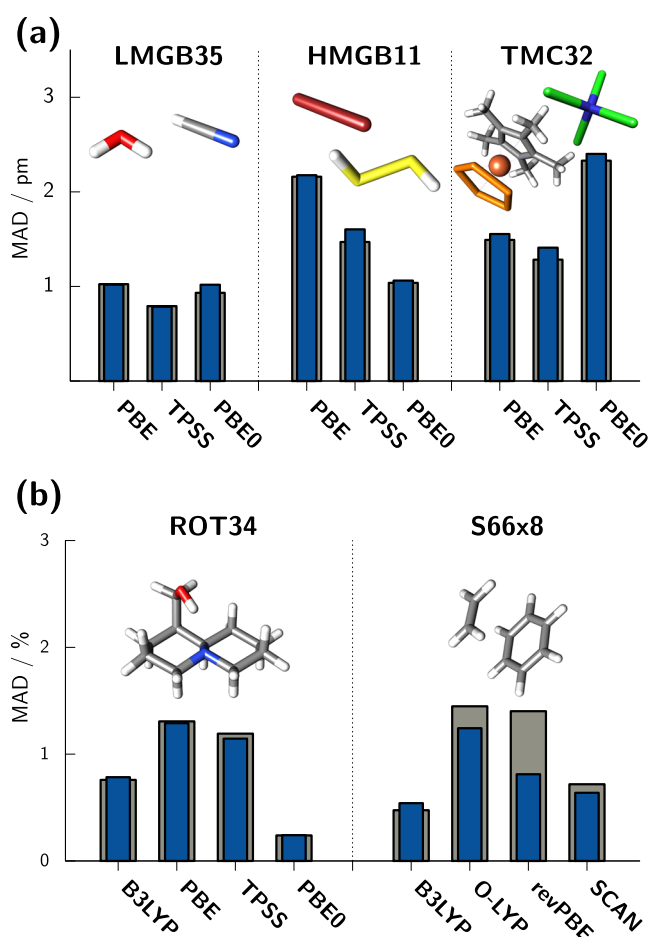


FIG. 10. (a) Mean absolute deviations from reference values for covalent bond lengths for the LMGB35, HMGB11, and TMC32 benchmark sets with three DFAs. Gray bars: DFT-D3(BJ)-ATM and blue bars: DFT-D4 (thinner). (b) Mean relative deviations from reference values for two benchmark sets representing the quality of entire equilibrium structures (rotational constants, ROT34) and noncovalent center-of-mass equilibrium distances (derived from S66x8) for seven DFAs. Gray bars: DFT-D3(BJ)-ATM (bar width 1.0) and blue bars: DFT-D4 (bar width 0.4).

noncovalent distance regime which is not really covered in these relatively small molecules tested. The tiny worsening observed for HMGB11 and for the TMC32 benchmark (the MAD increases by less than 0.1 pm) is practically irrelevant.

Small differences between DFT-D4 and DFT-D3(BJ)-ATM are found for equilibrium rotational constants B_e , while larger differences occur for intermolecular distances in the S66¹²⁸ noncovalent equilibrium complex benchmark (as derived from data for the S66x8 benchmark set⁷³). The B_e values can be considered as a measure for the quality of a complete molecular structure. Its accurate computation requires a consistently good description of covalent bond lengths as well as of nonbonded distances, and small changes of internal rotational degrees of freedom may result in rather large deviations of a few percent for B_e . As already discussed in the literature, dispersion corrections to DFT typically increase B_e values (shrink molecular size) significantly by about 0.5%–1.5%,¹²⁹ thereby in general improving the agreement with the reference data.

The left side of Fig. 10(b) shows the relative MAD in percentage of computed equilibrium rotational constants compared to back-corrected experimental values for the ROT34 benchmark set.¹³⁰ Here, dispersion corrected PBE0 yields accurate molecular structures within the uncertainty of the reference data while dispersion corrected TPSS and PBE on average overestimate molecular size (see the statistical evaluation in the [supplementary material](#)). Dispersion corrected B3LYP provides slightly enlarged molecular structures where both, the D4 and the D3(BJ)-ATM correction, perform equally well. All in all, the high quality of covalent equilibrium geometries obtained with DFT-D3(BJ)-ATM is retained with DFT-D4.

Noncovalent equilibrium geometries are considered on the right side of Fig. 10(b). Reference coupled-cluster interaction energies taken from Ref. 73 for eight different shifted center-of-mass (CMA) distances were interpolated to obtain reference CMA distances for all complexes in the S66 set. The DFT computed CMA distances were obtained in the same way to allow a direct comparison to the reference values.

As can be seen from this plot, the D4 model yields a slight (O-LYP and SCAN) to enormous improvement (revPBE) compared to the already very accurate DFT-D3(BJ)-ATM geometries. For the standard DFAs PBE, TPSS, and PBE0 shown in Fig. 10(a), D4 performs slightly better or equally well than D3-ATM. Note that the relative error for calculated CMA distances without dispersion correction is about one order of magnitude larger and that the very small residual relative deviations of 1.0% or less are similar for mostly covalent (ROT34) and noncovalent (S66) sets, respectively.

IV. SUMMARY AND CONCLUSIONS

We presented the theory and main features of the final D4 method, in which charge-scaled atom-in-molecule dipole polarizabilities are interpolated by means of an atomic coordination number within a Gaussian weighting procedure to generate charge- and geometry-dependent atom-in-molecule polarizabilities $\alpha(i\omega)$. The partial charges used in the polarizability scaling are obtained by default with a classical electronegativity equilibration (EEQ) model. As fall-back levels, TB based Mulliken-type partial charges q^{TB} (GFN2-xTB) or DFT based Hirshfeld partial charges q^{DFT}

(PBE0/def2-TZVP) are proposed. The polarizabilities are then numerically integrated at runtime of the D4 code to obtain system-dependent dipole-dipole dispersion coefficients C_6^{AB} . For two benchmark sets, they are more accurate than the charge-independent ones from the predecessor D3 scheme, and particularly, better coefficients are obtained for transition metal compounds. The two-body dispersion energy is computed in the commonly applied form as a sum over pair interactions including dipole-dipole and approximate dipole-quadrupole terms. A well-known many-body dispersion model based on coupled harmonic oscillators can be used for the determination of all higher-order dipole terms using the charge- and geometry-dependent $\alpha(i\omega)$ values. Exact analytical gradients including derivatives of the EEQ charges are available for the simplified default energy expression where the higher-order dipole-dipole interactions are truncated to the ATM term for computational efficiency. The final DFT-D4 default model uses the two-body and the ATM energy expression [cf. Eq. (28)] in structure optimizations. In addition, a single-point MBD correction is recommended as a sanity check for larger systems [cf. Eq. (29)]. The degree of empiricism in the *ad hoc* more physical D4 is slightly higher than in D3 with the same number of functional-dependent damping parameters (three) and a few additional global parameters for the charge dependency and coordination numbers.

The accuracy of the D4 method has been extensively verified using various benchmark sets including dispersion coefficients (organic and organometallic systems) and with various standard density functionals as DFT-D4 for interaction energies (intramolecular dispersion and host-guest systems of medium to large size), conformational energies (sugars, tri- and tetrapeptides, inorganic systems, and RNA backbone models), covalent bond lengths (light main group elements, heavy main group elements, and 3d-transition metal elements), organic equilibrium structures (rotational constants), noncovalent equilibrium structures, and reaction energies of main group chemistry (extended GMTKN55 database) and transition metal complexes occurring in homogeneous catalysis (MOR41).

DFT-D4 in general improves slightly upon the already accurate and well-established DFT-D3(BJ)-ATM method, and the new D4 model seems to be the most accurate and general molecular dispersion correction available. This mainly results from the inclusion of atomic charge effects into the atomic polarizabilities, the balanced treatment of higher-order dipole-dipole interactions, and most importantly the high accuracy of the underlying TD-DFT dispersion coefficients. In particular, systems involving large atomic charges or respective charge changes in chemical processes (e.g., oxidation/reduction) benefit from a DFT-D4 treatment compared to DFT-D3. Moreover, it should be emphasized that the generally good description of dispersion interactions in nonpolar main group or organic molecules is retained compared to DFT-D3(BJ)-ATM, i.e., DFT-D4 represents a save replacement of a widely used and well tested method. In the current form, the D4 method appears to have reached a plateau level within the DFT-D methodology in terms of physical sophistication and cost-accuracy ratio. Further improvements would most likely lead to significantly increased computational costs and moreover would require concomitant improvements of the applied DFAs or damping schemes. Importantly, the general black-box philosophy of the DFT-D4 approach and the coupling of a single dispersion model to various standard DFAs are

analogous to DFT-D3. This and the availability for most of the periodic table (elements up to atomic number $Z = 86$) enable broad application to a wide variety of quantum chemical problems. A list of all 67 currently parameterized DFAs together with their BJ-damping parameters is given in the [supplementary material](#) (for different atomic partial charges in combination with an ATM or an MBD treatment).

The proposal of D4 with a robust and reasonably accurate classical charge model as basis allows its application not only in the DFT context. Simplified semi-empirical orbital models, sophisticated force-fields¹³¹ or neural network potentials can easily be coupled with the D4 model simply because no electron density is required and the whole procedure including nuclear gradients still runs close to “force-field speed.” This opens a route for an efficient, physically sound, and basically nonempirical treatment of noncovalent interactions in large systems with thousands of atoms.

Ongoing work focuses on the implementation of the D4 model for periodic systems, which requires a cost-efficient determination of partial charges under periodic boundary conditions. The here presented final molecular (nonperiodic) version can be downloaded in coded form from the authors’ website free of charge.¹³² Furthermore, the DFT-D4 method is available in the upcoming release of TURBOMOLE 7.3 software,^{43–45} the next release of ORCA 4.1.0,⁶⁷ and within an update of MOLPRO.¹³³

SUPPLEMENTARY MATERIAL

In the [supplementary material](#), the classical charge model is defined in more detail including an energy expression and analytical gradients. The theoretical foundations of the many-body dispersion correction used within the DFT-D4-MBD method are discussed in detail. The derivation of the two-body dispersion potential which is included into the GFN2-xTB Hamiltonian matrix is given. The definitions of the double hybrid density functionals used are given. The BJ-damping parameters are given for 67 DFAs and for Hartree-Fock. We compare timings (single-point energies and gradients) between DFT-D4 and DFT-D3(BJ)-ATM for the tetrakis(isonitrile)rhodium(I) dimer with 106 atoms and for a diamond chunk with 430 atoms. Furthermore, all statistical quantities are defined and listed as used in the evaluation of the calculated data. Statistical evaluations of the following benchmark systems are given with all reference and computed data for S30L, L7, MOR41, SCONF, PCONF21, ICONF, UPU23, ROT34, LMGB35, HMGB11, and TMC32. Structures of the tetrakis(isonitrile)rhodium(I) monomer and dimer are attached, whereas the TOMC6 benchmark set (structures and reference molecular C_6 coefficients) is given as a tarball. Furthermore, we share the fitting set used for determination of damping parameters to ensure that new DFAs can be parameterized consistently.

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